

2022 AIME I Problems/Problem 10

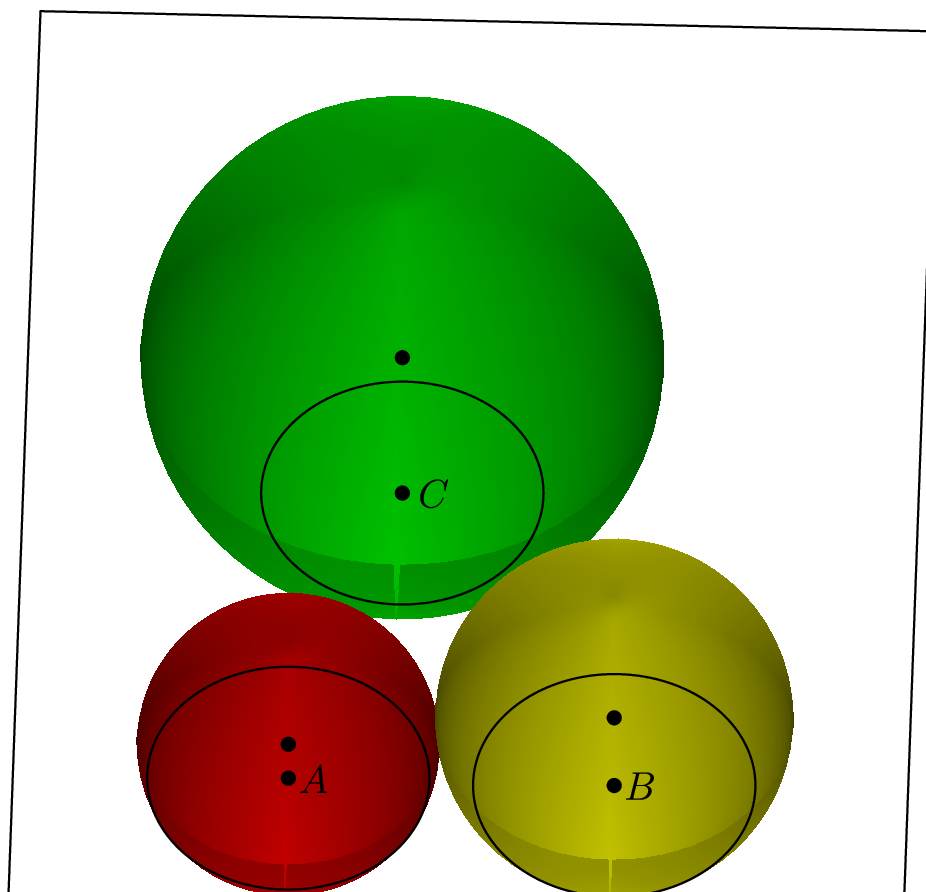
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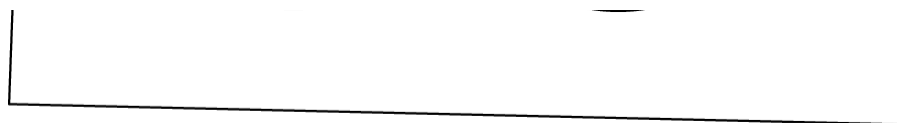
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Problem

Three spheres with radii 11 , 13 , and 19 are mutually externally tangent. A plane intersects the spheres in three congruent circles centered at A , B , and C , respectively, and the centers of the spheres all lie on the same side of this plane. Suppose that $AB^2 = 560$. Find AC^2 .

Diagram



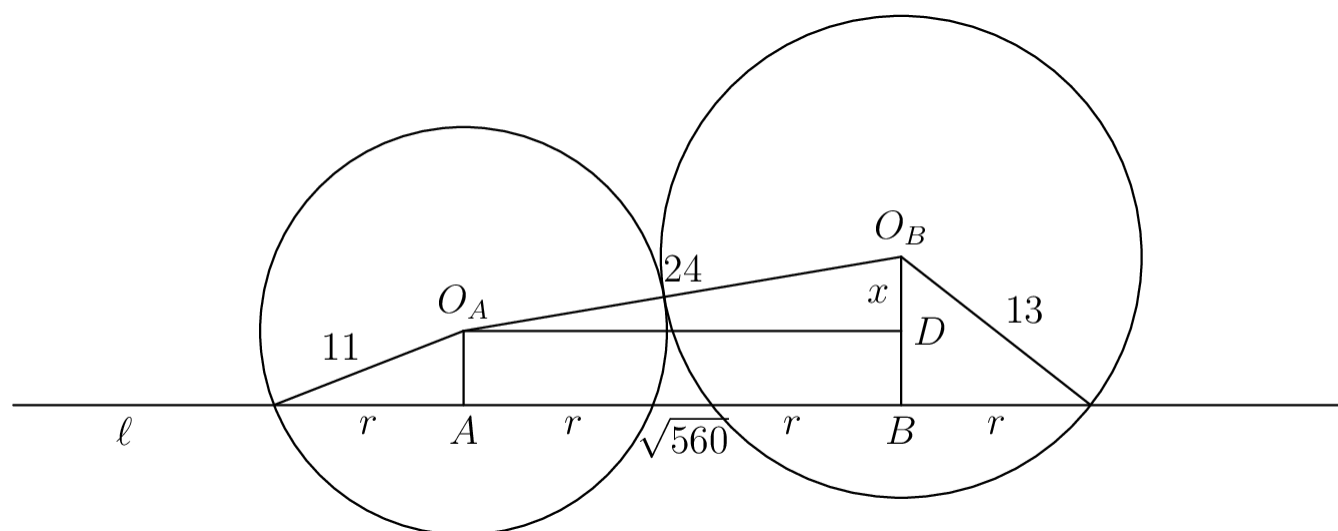


~MRENTHUSIASM

Solution 1

This solution refers to the **Diagram** section.

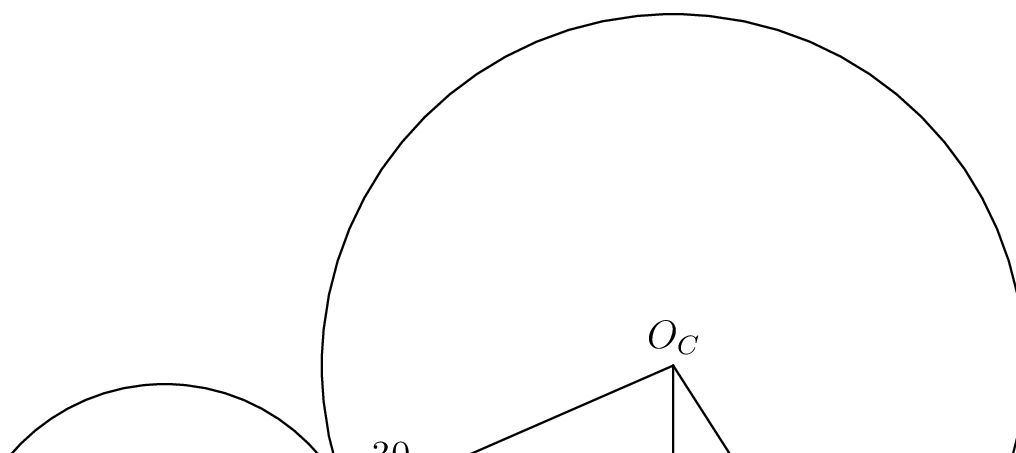
We let ℓ be the plane that passes through the spheres and O_A and O_B be the centers of the spheres with radii 11 and 13. We take a cross-section that contains A and B , which contains these two spheres but not the third, as shown below:

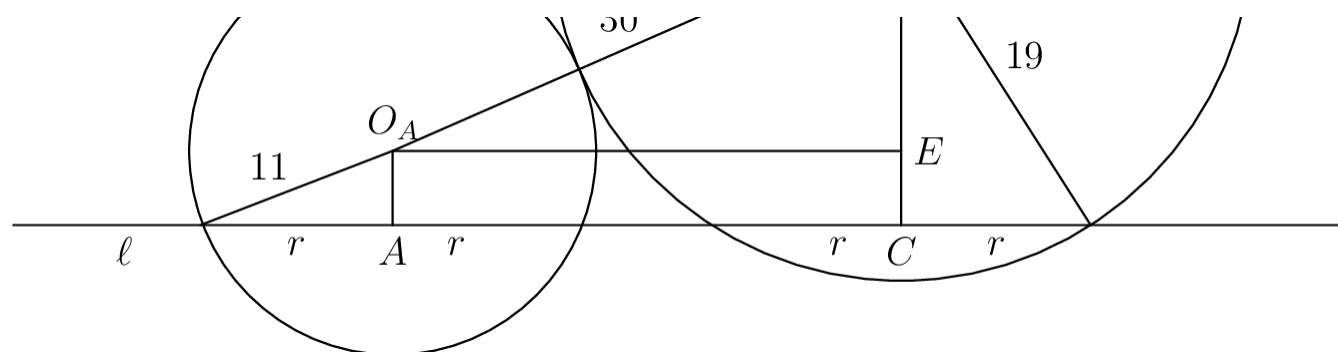


Because the plane cuts out congruent circles, they have the same radius and from the given information, $AB = \sqrt{560}$. Since ABO_BO_A is a trapezoid, we can drop an altitude from O_A to BO_B to create a rectangle and triangle to use Pythagorean theorem. We know that the length of the altitude is $\sqrt{560}$ and let the distance from O_B to D be x . Then we have $x^2 = 576 - 560 \implies x = 4$.

We have $AO_A = BD$ because of the rectangle, so $\sqrt{11^2 - r^2} = \sqrt{13^2 - r^2} - 4$. Squaring, we have $121 - r^2 = 169 - r^2 + 16 - 8 \cdot \sqrt{169 - r^2}$. Subtracting, we get $8 \cdot \sqrt{169 - r^2} = 64 \implies \sqrt{169 - r^2} = 8 \implies 169 - r^2 = 64 \implies r^2 = 105$. We also notice that since we had $\sqrt{169 - r^2} = 8$ means that $BO_B = 8$ and since we know that $x = 4$, $AO_A = 4$.

We take a cross-section that contains A and C , which contains these two spheres but not the third, as shown below:





We have $CO_C = \sqrt{19^2 - r^2} = \sqrt{361 - 105} = \sqrt{256} = 16$. Since $AO_A = 4$, we have $EO_C = 16 - 4 = 12$. Using Pythagorean theorem, $O_A E = \sqrt{30^2 - 12^2} = \sqrt{900 - 144} = \sqrt{756}$. Therefore, $O_A E^2 = AC^2 = \boxed{756}$.

~KingRavi

Solution 2

Let the distance between the center of the sphere to the center of those circular intersections as a, b, c separately.

According to the problem, we have $a^2 - 11^2 = b^2 - 13^2 = c^2 - 19^2$; $(11 + 13)^2 - (b - a)^2 = 560$. After solving we have $b - a = 4$, plug this back to $11^2 - a^2 = 13^2 - b^2$, we have $a = 4, b = 8$, and $c = 16$.

The desired value is $(11 + 19)^2 - (16 - 4)^2 = \boxed{756}$.

~bluesoul

Solution 3

Denote by r the radius of three congruent circles formed by the cutting plane. Denote by O_A, O_B, O_C the centers of three spheres that intersect the plane to get circles centered at A, B, C , respectively.

Because three spheres are mutually tangent, $O_A O_B = 11 + 13 = 24$, $O_A O_C = 11 + 19 = 30$.

We have $O_AA^2 = 11^2 - r^2, O_BB^2 = 13^2 - r^2, O_CC^2 = 19^2 - r^2$.

Because $O_A A$ and $O_B B$ are perpendicular to the plane, $O_A A B O_B$ is a right trapezoid, with $\angle O_A A B = \angle O_B B A = 90^\circ$.

Hence,

$$\begin{aligned} O_B B - O_A A &= \sqrt{O_A O_B^2 - AB^2} \\ &= 4. \end{aligned} \quad (1)$$

Recall that

$$\begin{aligned} O_B B^2 - O_A A^2 &= (13^2 - r^2) - (11^2 - r^2) \\ &= 48. \end{aligned} \quad (2)$$

Hence, taking $\frac{(2)}{(1)}$, we get

$$O_BB + O_AA = 12. \quad (3)$$

Solving (1) and (3), we get $O_BB = 8$ and $O_AA = 4$.

Thus, $r^2 = 11^2 - O_AA^2 = 105$.

Thus, $O_CC = \sqrt{19^2 - r^2} = 16$.

Because O_AA and O_CC are perpendicular to the plane, O_AACO_C is a right trapezoid, with $\angle O_AAC = \angle O_CCA = 90^\circ$.

Therefore,

$$\begin{aligned} AC^2 &= O_AO_C^2 - (O_CC - O_AA)^2 \\ &= \boxed{756}. \end{aligned}$$

FINAL NOTE: In our solution, we do not use the condition that spheres A and B are externally tangent. This condition is redundant in solving this problem.

MORE FINAL NOTE: the above note is incorrect because that condition was used at the start when claiming $O_AO_B = 24$. Perhaps the note is referring to spheres B and C .

~Steven Chen (www.professorchen.eu.com)

~anonymous (minor edits)

Video Solution 1 by Challenge 25

<https://www.youtube.com/watch?v=yeuJDQ1LTIY>

Video Solution 2

<https://www.youtube.com/watch?v=SqLiV2pbCpY&t=15s>

~Steven Chen (www.professorchen.eu.com)

Video Solution 3 Mathematical Dexterity

<https://www.youtube.com/watch?v=HbBU13YiopU>

Video Solution 4 by StressedPineapple

<https://youtube.com/watch?v=x66z0IMj0as&t=303s>

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